PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for B.voc (Nutrition &Dietetics)

as per NEP 2020

w.e.f session 2026-27

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as per NEP 2020 w.e.f. session 2026-27

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 101	Core Discipline	Human Physiology	2	2	4	40	10	40	10	100
NAD 102	Core Discipline	Basics of food and nutrition	2	2	4	40	10	40	10	100
NAD 103	Core Discipline	Indigenous Food systems	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	Common Basket			3					75
IDC I	Interdisciplinary Course	Common Basket			3					75
VAC I	Value-Added Course	Common Basket			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 201	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
NAD 202	Core Discipline	Food Preservation	2	2	4	40	10	40	10	100
NAD_203	Core Discipline	Nutrition and Immunity	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	Common Basket			3					75
IDC II	Interdisciplinary Course	Common Basket			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role –Food Microbiologist

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter2ndSemesterExaminations during summer break. Training report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rdsemester.
** Minimum Credits required to get Certificate Course in Nutrition & Dietetics = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a

UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 301	Core Discipline	Nutritional Biochemistry	2	2	4	40	10	40	10	100
_NAD 302	Core Discipline	Nutraceutical and Food additives	2	2	4	40	10	40	10	100
_NAD 303	Core Discipline	Food science	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC III	Skill Enhancement Course	Common Basket			3					75
IDC III	Interdisciplinary Course	Common Basket			3					75
VAC II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 401	Core Discipline	Human Nutrition	4	-	4	80	20	-	-	100
_NAD 402	Core Discipline	Institutional Food Service & Personnel Management	2	2	4	40	10	40	10	100
_NAD 403	Core Discipline	Meal management	2	2	4	40	10	40	10	100
NAD 404	Core Discipline	Food Quality Management	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role –Nutrition educator

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter4th Semester Examinations during summer break. Training report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rdsemester.

** Minimum Credits Required to get Diploma Course in Nutrition & Dietetics = 96 + 4 #.

#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 501	Core Discipline	Therapeutic Nutrition-I	2	2	4	40	10	40	10	100
NAD 502	Core Discipline	Food Hygiene & Sanitation	4	-	4	80	20	-	-	100
NAD 503	Core Discipline	Food standards & Quality Control	4	-	4	80	20	-	-	100
NAD 504^	Core Discipline	Community Nutrition-I	2	2	4*	40	10	40	10	100
NAD 505^	Core Discipline	Health, Fitness and Sports Nutrition	2	2	4*	40	10	40	10	100
NAD 506^	Core Discipline	Diet Counselling	4	-	4*	80	20	-	-	100
VAC IV	Value-Added Course	Common Basket			2					50
PW III	Project Work	Project Work-III& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 601	Core Discipline	Therapeutic Nutrition II	2	2	4	40	10	40	10	100
NAD 602	Core Discipline	Introduction to Research Methodology and Report Writing	4	-	4	80	20	-	-	100
NAD 603	Core Discipline	Total Quality Management	4	-	4	80	20	-	-	100
NAD 604	Core Discipline	Seminar I	-	4	4	-	-	80	20	100
NAD 605^	Core Discipline	Community Nutrition-II	2	2	4*	40	10	40	10	100*
NAD 606^	Core Discipline	Clinical Assessment and Patient Counselling	2	2	4*	40	10	40	10	100*

NAD 607^	Core Discipline	Maternal & Child Health and Nutrition	4	-	4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 6th Semester

Job Role – Clinical diet counselor

Semester-VII

Paper Code	**Type of Course**	**Subject Name**	**Credits**			**Marks**				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 701	Core Discipline	Advance nutrition	4	-	4	80	20	-	-	100
NAD 702	Core Discipline	Dietetics and Therapeutic Nutrition	2	2	4	40	10	40	10	100
NAD 703	Core Discipline	Seminar II	-	4	4	-	-	80	20	100
NAD 704^	Core Discipline	Public Health and Nutrition	2	2	4*	40	10	40	10	100*
NAD 705 ^	Core Discipline	Principles of food processing	2	2	4*	40	10	40	10	100*
NAD 706 ^	Core Discipline	Institutional Food Administration	4	-	4*	80	20	-	-	100*
SIT 701	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

^Any two in 7th Semester

Semester-VIII

Paper Code	**Type of Course**	**Subject Name**	**Credits**			**Marks**				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 801	Core Discipline	Advanced Nutritional Biochemistry	2	2	4	40	10	40	10	100
_NAD 802	Core Discipline	Clinical Nutrition	2	2	4	40	10	40	10	100

NAD 803	Core Discipline	Women Health and Hygiene	4	-	4	80	20	-	-	100
NAD 804	Core Discipline	Dissertation	-	4	4	-	-	80	20	100
NAD 805^	Core Discipline	Extension Education in Nutrition	4	-	4*	80	20	-	-	100*
_NAD 806^	Core Discipline	Techniques in food analysis	2	2	4*	40	10	40	10	100*
NAD 807 ^	Core Discipline	Food service management	4	-	4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 8th Semester

Job Role –Human Resource Officer

* SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter6th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 7th Semester. Viva-Voce examination to be held within 3 weeks of the start of 7thsemester.

PROGRAMMEOUTCOMES

PO1. Prepare qualified entry level professionals by providing expected domain knowledge.

PO2. Prepare a socially committed individual having high ethical values and empathy for the needs of society.

PO3. Design/development of solutions Develop innovative food products or substitutes or alternate solutions to create value and wealth for the betterment of the individual and society at large.

PO4. Problem analysis Identify, formulate, research literature, and solve nutritional deficiencies using fundamentals of clinical nutrition and dietetics, physiology, food science and biochemistry and relevant domain disciplines.

PO5. Professional Ethics Hold up and commit to professional ethics and ethical regulations, responsibilities, and norms of professional nutrition and dietetics practice.

PO6. Graduates are trained to develop innovative ideas and teams work skills to accomplish a common goal.

PO7. Produce lifelong learner with scientific skills for industrial applications and entrepreneurship.

Semester I

1	Course Title	Human Physiology
2	**Course Code**	**NAD 101**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Exercise Physiologist
8	**Course Objectives**	1. To provide a basic introduction to human physiology: the study of the functions of the human body and its constituent parts. **2.** Apply knowledge of functional mechanisms and their regulation to explain the underlying physiology of common disease.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. This course is based on basics of Anatomy and Physiology highlighting structural anatomy of various human body organs. CO2. To understand how these separate systems interact to yield integrated physiological responses. CO3. Students will be able to associate the structural anatomy with functions of body organs. CO4. The focus of this course is to evaluate the effect of alterations in structure on the functions of organs. CO5. The students under this course will able to understand the working of equipment's and instruments used in physiology lab. CO6. The course gives an insight about various parameters for diagnosis of illness.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

UNIT– I

• Cell structure: Function and multiplication. Introduction to cell physiology and cell membrane

- Circulatory system: Anatomy and functions. Heart rate, cardiac cycle. Cardiac output. Blood composition and functions. Blood clotting, blood groups, radial pulse, blood pressure and factors affecting it.

UNIT– II

• Digestive system: Anatomy and functions of alimentary tract and accessory organs. Digestion and absorption of food.

- Respiratory system: Anatomy, functions, mechanisms of external and internal respiration

UNIT– III

• Endocrine System: List of endocrine glands, their hormones, functions, and their disorders (in brief)- Thyroid, adrenal, pancreas.

• Urinary system: Anatomy and functions. Formation and composition of urine.

UNIT-IV

• Skin: Structure and functions

• Sensory organs: Eye, Ear, Nose, Tongue, skin – their function and basic structure (in brief).

Practical work:

1. Study of different types of microscopes and their uses.
2. Microscopic examination of permanent slides.
3. Identification of different blood groups.
4. Measurement of pulse and blood pressure.
5. Estimation of haemoglobin, RBC, WBC, TLC, DLC and ESR.

13 Suggested Resources for Reading

1. Keele, C.A. and Neil, E. Samson Wright 's Applied Physiology, Oxford University Press.
2. Tortora, G.J. and Anagnostakos, N.P. (1984). Principles of Anatomy and Physiology, Harper and Row Publisher.
3. Best, C.H. and Taylor, N.B., The Living Body, Asia Publishing House
4. Guyton, A.C., Functions of the Human Body, W.B. Saunders Co.
5. Wilson, K.J.W., Anatomy and Physiology in Health and Illness, ELBS/Churchill Livingstone.
6. Vidhya Ratan Handbook of Human Physiology, New Central Book Agency

Note: Latest edition of text books may be used.

1	**Course Title**	**Basics of Food and Nutrition**
2	**Course Code**	**NAD 102**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutritionist
8	**Course Objectives**	1. To elucidate the applied concepts of different areas of Foods and Nutrition. **2.** To acquire knowledge about the nutritional needs and concerns of specific age groups/physiological conditions.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. The course focuses on the functions of food for healthy life. CO2. To designed to make the students understand different aspects of nutrients. CO3.To designed to make the students understand different aspects of nutrients. CO4. The students will be able to understand the nutrition information based on scientific reasoning for clinical, community, and food service applications. CO5. This practical course is designed make the students understand the importance of weights and measures in nutritional cookery. CO6. It is designed to know the nutritional value of basic recipes of each category through calculations.
10	**Pedagogy of the Course Work**	**Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.**
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Introduction to nutrition: Food as a source of nutrients, food pyramid, function of foods, definition of nutrition, nutrients, and energy; adequate, optimum and good nutrition, malnutrition.
- Food guide – Basic five food groups and the nutritional contribution of each group in the diet.
- Interrelationship between nutrition & health -Visible symptoms of good health

UNIT– II

- Carbohydrates – Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body. Role of fibre in human nutrition.
- Fats & Oils - Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body.

UNIT– III

- Proteins- Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body.
- Water–role as a nutrient, functions, sources, requirement, water balance & effect of deficiency.
- Vitamins – their functions, sources, requirements, effect of deficiency and excess of following vitamins-
  a) Fat Soluble - Vitamins A, D, E and K
  b) Water Soluble- Ascorbic acid and B-Complex: Thiamine, Riboflavin, Niacin, Folic acid and B12.

UNIT-IV

- Minerals –their functions and sources. Bioavailability, requirements, and effect of excess and deficiency of calcium, iron, iodine and sodium.
- Functional foods.
- Importance and methods of cooking and their effect on the appearance, nutritional value & taste**.**
- **Practical work:**

1. Use and Care of different Kitchen equipment's
2. Weights and measures-Standard, Household measures for raw and cooked food.
3. Preparations from cereals (ANY TWO)- e.g. boiled rice, rice, pulao, chapati, puri, paratha, sandwiches, pastas, pancakes, cookies, and cake.
4. Preparations from pulses (ANY TWO)- e.g. dal makhani, sambar, chana masala, rajma etc.
5. Preparations from vegetables (dry, curries, koftas) (ANY TWO)- e.g. mixed vegetable, gobhi masala, vegetable kofta etc.
6. Preparations from milk (ANY TWO)- e.g. Porridge, curd/raita, paneer dishes, desserts and puddings custard, kheer, ice cream.
7. Soups (ANY TWO)- Basic, clear, and cream soups.
8. Salads with and without dressing. (ANY TWO)
9. Beverages (ANY TWO)- preparation of nourishing and refreshing beverages.
10. Snacks (ANY TWO) – pakoras, cutlets, cheese toast, upma, poha, peanut chikki, til-laddu preparations.
11. Use of leftover foods.

Requirements of all the nutrients should be according to RDA by NIN, ICMR

13 Suggested Resources for Reading

1. Guthrie, H., Andrews, Introductory Nutrition, St. Louis, Mirror /Mosby Collage.
2. Mudami S.R. &M. V Rajagopal, Fundamentals of Food and Nutrition, Wiley Eastern
3. Swaminathan M., Advanced Textbook of Foods, and Nutrition, Vol I, II, Bappco, Banga lore.
4. Wilson, E.D., Principals of Nutrition, John Wiley &Sons
5. Swaminathan, M, Principals of Nutrition and Dietetics, Bangalore Print & Publishing Co.
6. William, S., Nutrition and Diet Therapy, Mosby Co, St. Louis.
7. Bogert, L.J., & Calloway, D.H., Nutrition and Physical Fitness, W.B. Saunders & Co, Philadelphia.

Note: Latest edition of text books may be used.

1	**Course Title**	**Indigenous Food System**
2	**Course Code**	**NAD 103**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80THEORY+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Ayurvedic physicians
8	**Course Objectives**	1. To impart knowledge regarding promotion and conservation of medicinal plants **2.** Apply knowledge regarding Ayurvedic medicinal plants and its impacts on health.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. This course describes both broad knowledge and in-depth relation of herbs and nutrition. CO2. It is based on study of different herbs and their use in day-to-day life. CO3. The course also relates nutrition and functional components of herbs by exploring the newer product development.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Ayurveda: Concept, history, classification and uses
- Basic principles of Ayurveda, Allied systems of medicine in
- Ayurveda. Concept of diet in Ayurveda.
- Status of Ayurveda and Perspectives for its future application.

UNIT– II

- Introduction to medicinal plants- Aloe Vera, Tulsi, Neem, Turmeric, Ginger, Moringa, Amalta and Brahmi; functional properties, traditional practices and nutritional significance Promoting sustainability, conservation, and utilization of medicinal plants.

UNIT– III

- Role of herbs and spices in developed and developing countries.
- Utility of medicinal plants in household remedies of curing various diseases
- Functional Foods grown at household level and its role on human health; Factors influencing use of food for medicinal purposes.

UNIT-IV

- Medicinal Plant market and industries in India.
- Medicinal food products of the future - novel foods

13 Suggested Resources for Reading

1. Joshi, M.C., Text book of Indian Medicinal Plants –Scientific Publishers
2. Bhutya, R K., Handbook of Ayurvedic Medicinal Plants of India-Scientific Publishers
3. Kapoor, L.D., Handbook of Ayurvedic Medicinal Plants

Note: Latest edition of text books may be used.

SEMESTER – I

PW I
Project Work

- Survey to evaluate the dietary habits and assess the nutritional status of adolescent girls and middle-aged women, followed by an analysis and recommendation.
- Proposing healthy and nutritious recipes tailored for these demographic group.
- Submitting of project report.

Semester II

1	Course Title	Food Microbiology
2	**Course Code**	**NAD 201**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80THEORY+10IA) + (40 PRACTICAL + 10 IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not by provided by the University/College to the examinees
7	**Job Role**	Food Microbiologist
8	**Course Objectives**	1. To study the origin of micro-organisms and their structure. 2. To understand the growth and activities of bacteria. 3. To gain knowledge of principles of various techniques used in prevention and control of micro- organisms in food. **4.** To develop the knowledge about harmful bacteria, methods of transmission and anti-microbiology agents: antibiotics, germicides
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Explain the interactions between microorganisms and the food environment, and factors influencing their growth and survival. CO2. Explain the significance and activities of microorganisms in food. CO3. Describe the characteristics of foodborne, waterborne and spoilage microorganisms, and methods for their isolation, detection, and identification. CO4. Explain why microbiological quality control programme are necessary in food production.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

UNIT– I

Introduction of microbiology and its relevance to everyday life, general characteristics of bacteria, fungi, virus and algae.

Growth of microorganisms: Growth curve, effect of environmental factors in growth of microorganism, pH, water activity, oxygen availability and temperature.

UNIT– II

Microbiology of different foods: Spoilage and contamination, sources, types, effect on the following:

(a) Cereal and cereal products
(b) Sugar and sugar products
(c) Vegetables and fruits
(d) Meat and meat products
(e) Milk and milk products

UNIT– III

Environmental microbiology:

(a) Water and water borne diseases
(b) Air and air borne diseases
(c) Soil and soil borne disease

Beneficial effect of microorganisms.

Fermented foods: Bread, malt beverages.

UNIT– IV

Role of microorganisms in fermented milk products: Butter and cheese etc.

Microbial enzymes in food processing. Relevance of microbial standards for food safety.

Microbial intoxication and infections: Sources of contamination of food by pathogenic organisms, symptoms, and methods of control.

Practical work:

1. Study of equipment's in a microbiology lab.
2. Preparation of laboratory media and special media, cultivation and identification of bacteria, yeasts, and moulds.
3. Gram staining of bacteria.
4. Microbial examination of various foods: Foods and vegetables, meat, milk, and milk products.
5. Dye reduction test for assessing quality of milk.
6. Sterilization of various utensils.
7. Testing of water quality.
8. Development of aseptic conditions.
9. Visits to food processing units or any other organization dealing with advanced methods in food microbiology.

13 Suggested Resources for Reading

1. Frailer, W.C., Food Microbiology, McGraw Hill, New York
2. Kawata K. Environmental Sanitation in India, Lucknow Publ. House
3. Pelezer, H.J. and Rober,D. Microbiology, McGraw Hill, New York
4. Banwart, G.T., Basic Food Microbiology, CBS Publ., New Delhi
5. Jay,J.H, Modern Microbiology, CBS Publ., New Delhi

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Preservation and Storage**
2	**Course Code**	**NAD 202**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY+10IA) + (40 PRACTICAL + 10 IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food technologists
8	**Course Objectives**	1. To impart knowledge on the causes of food spoilage and methods of processing and preserving food. 2. To identify & select processing equipment and preservation methods appropriate for specific foods. **3.** To describe the effects of methods of preservation on the quality of food.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. To understand the basic principles of food preservation CO2. To discuss the qualities and standards of available food preservatives and their suitability for different purposes. CO3. In this portion students learn about various techniques of preserving food products and detailed study of preserved food products, storage, nutritional aspects and food standards and food laws. CO4. students learn different methods of Food preservation – Drying, Freezing, Frying, canning, bottling etc. and preparation of jam, jelly, pickles, sauces, squashes.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Definition, scope, and importance of food preservation.
- Classification of food based on its shelf life.
- Food spoilage: cause and effects.
- Principles of Food preservation
- Techniques of food preservation

UNIT– II

- Methods of food preservation-Canning: Principles of Preservation of canned foods.
- Water activity of food and its significance in food preservation
- Dehydration and drying of food items
- Low temperature preservation: cold storage, cold chain, freezing (including cryogenic freezing)
- Preservation by fermentation: curing and pickling; Use of preservative in foods
- Chemical preservative, bio preservatives, antibiotics, lactic acid bacteria

UNIT– III

- Selection and purchase of foods for preservation.
- Storage of common perishable, semi-perishable and non-perishable foods.
- Storage stability of selected Frozen Foods.

UNIT-IV

- Packaging and packing materials, labelling, and costing of products.
- Food additives, preservatives, anti-oxidants, colouring matter, and emulsifying agents.

Practical work:

1. Preparation and storage of:
    a) Squashes and syrups
    b) Jams and jellies
    c) Ketchup/sauces and chutneys
    d) Pickles (with and without oil).
2. Preparation of items utilizing cereals and legumes and their storage.
3. Demonstration of canning and bottling of fruits and vegetables.
4. Demonstration of Sealing Process used in Food Industry
5. Visit to food industry.

13 Suggested Resources for Reading

1. Desrosier, L.P, The Technology of Food Preservation, CBS Publishers & Distributors
2. Potter, N.N & Hotchkiss, J.H., Food Science, CBS Publishers & Distributors
3. Srivastava, R.P. & Kumar, S., Fruits and Vegetable Preservation, Principle and Practices, CBS Publishers & Distributors
4. Muller, H.G., Fundamentals of Foods and Nutrition. Wiley Eastern Pvt Ltd.
5. Modern Technology on Food preservation by NIIR Board.

Note: Latest edition of text books may be used.

1	**Course Title**	**Nutrition & Immunity**
2	**Course Code**	**NAD 203**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80THEORY+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutrition consultant
8	**Course Objectives**	Examining how nutrition reduces the side effects of medical treatments in general.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. To provide knowledge about various foods and their quality and how to use them to provide immunity to the human body. CO2. The immune system can boost some components, such as vitamins, minerals, antioxidants, and phenolic compounds, because these components aid in forming T cells, B cells, and antibodies and enhance the function of melanoproteins and neutrophils. CO3. To understand the role of functional food ingredients of various foods and their mechanism. CO4. To illustrate the health benefits of using nutraceuticals
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Definition of immunity, Immune system: The defence mechanisms of the body.
- Role of nutrients in immune functions. Under nutrition-causes and consequences.

UNIT– II

•Infection: Risk of infection, influence of infections on nutrition and susceptibility to infection.
• Role of nutrients in prevention of risk of infections.
• Childhood mortality associated with infectious diseases of gastro-intestinal tract and lungs.

UNIT– III

• Age related immune depression.
• Role of nutraceuticals and functional foods.
• Nutrition in HIV/AIDS and tuberculosis.

UNIT-IV

• Probiotics, prebiotics, and immunity
• Food allergy (with emphasis on wheat allergy, peanut, lactose intolerance)
- Latest health concerns

13 Suggested Resources for Reading

1. Barasi, M. E., Human Nutrition-A health perspective. Arnold International Student ‘s edition.
2. Gershwin, M.E., Nestel, P. and Keen, C.L., Handbook of Nutrition and Immunity. Humana Publishers
3. Owen, J., Punt, J. and Stanford, S., Kuby Immunology: International Edition. Macmillan Publishers.
4. Prason, S.K. and Poddar, T., Perfect Health-Body Diet and Nutrition. V & S publishers.

Note: Latest edition of text books may be used.

SEMESTER III

1		**Nutritional Biochemistry**
2	**Course Code**	**NAD 301**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutritional Biochemist
8	**Course Objectives**	To lay the foundation for an introduction to various bio-molecules and understanding of various metabolic pathways**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to CO1**: Compare different metabolic pathways of carbohydrates. **CO2**: Distinguish between different metabolic pathways of lipid and its derivatives. **CO3**: Identify the structure and general pathways of amino acids metabolism **CO4**: Illustrate the integration of metabolic pathways of energy metabolism
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 **Course Content**

UNIT I

• Biochemistry, its scope and importance in foods and nutrition.
• Carbohydrates: Digestion, absorption and metabolism of carbohydrates. Anaerobic and aerobic pathways, glycolysis, citric acid cycle, gluconeogenesis, glycogenesis, glucose transport, metabolism of lactate and pyruvate (in brief).

UNIT II

• Fats and Lipids: Digestion, absorption and metabolism of fats and fatty acids, beta-oxidation of fatty acids, energy yield from dietary fats, storage, mobilization of fat stores during exercise, production of ketone bodies and ketogenic diets, metabolism of phospholipids (in brief).
• Proteins: Digestion, absorption and metabolism, urea cycle (in brief).

UNIT III

• Energy: BMR, Factors affecting BMR, energy requirements for physical activity, relative body weight and influence of physical exercise on changes in body fat and body composition, utilization of energy by muscle tissue in relation to exercise type, intensity and duration.
• Hormones: Classification, general mode of action, hormones of pituitary, thyroid, parathyroid, adrenals, reproductive glands and pancreas.

UNIT IV

• Enzymes: Definition, classification, co-enzymes, their functions and factors affecting enzyme activity.
• Acid base balance: Concepts and disorders, pH, buffers, acidosis, alkalosis.
• Water and electrolyte balance.

Practical Work

1. Preparation of standards solutions Measurement of pH.
2. Carbohydrates: Quantitative estimation of sugars in foodstuff.
3. Proteins: (i) Quantitative estimation of amino acids by Ninhydrin Method. (ii) Estimation of proteins by Lowry method and Biuret method (iii) Estimation of nitrogen by Khedjals Method
4. Lipids: (i) Quantitative estimation of cholesterol (ii) Isolation and estimation of total lipids (iii) Quantitative estimation of Phospholipids.

13 **Suggested Resources for Reading**

1. Rao, K.R., Textbook of Biochemistry, Prentice Hall of India Pvt. Ltd. ND- 110001.
2. Murray, R.K. and Rodwell, V.W. Harper's Biochemistry, Prentice Hall of Australia Pvt. Ltd.
3. Lehninger, A.L., Principles of Bio-Chemistry, CBS Publishers and Distribution.
4. Qrten, J.M. and Neuhans, O.W., Human Bio-Chemistry, The C.V. Mosby Co., Toronto, London.

Note: Latest edition of text books may be used.

1		**Nutraceuticals and Food Additives**
2	**Course Code**	**NAD 302**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Product Development Manager
8	**Course Objectives**	To learn about food components and their nutraceutical Importance. To learn the role of food additives
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** learn some important food components and their nutraceutical importance **CO2.** Understand the mechanism of action of some phytochemicals and zoochemical in the treatment and prevention of diseases such as cancer, cardiovascular diseases and so on. **CO3.** Know Indian foods and some of their claimed nutraceutical properties. **CO4.** Learn the role of food additives in processing food
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	5. **The syllabus of this paper has been divided into FOUR units.** 6. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 7. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 8. **All questions carry equal marks.**

12 Course Content

Unit I

- Nutraceuticals: Definition, Classification, food and non-food sources, mechanism of action. Role of omega-3, fatty acids, carotenoids, dietary fiber, phytoestrogens; gluconates,organo-Sulphur compounds as nutraceuticals. Use of nutraceuticals in traditional health sciences. Their role in preventing /controlling diseases.

Unit II

- Prebiotics and Probiotics: Usefulness of probiotics and prebiotics in gastrointestinal health and other benefits. Beneficiary microbes; prebiotic ingredients in foods; types of prebiotics and their effects on gut microbes.

Unit III

- Definitions, nutritional and non-nutritional food additives, uses and functions of Acid, Base, Buffer systems, Salts and chelating / sequestering agents.

Unit IV

- Food colour, pigments, their importance and utilization, Flavoring agents and related substances
- Antioxidants, Emulsifying and stabilizing agents, Anti-caking agents, Humectants, thickeners, Firming agents., Clarifying agents.

Practical Work

1. Identification of various nutraceuticals and functional foods available in the market
2. Preparation and sensory evaluation of probiotic/prebiotic/symbiotic foods
3. Preparation and sensory evaluation of antioxidant dietary fiber rich foods.
4. To conduct the market survey for identification of health claims of various nutraceuticals products.
5. Preparations of some traditional, fermented, functional and other products.

13 Suggested Resources for Reading

1. Colleen C., Kerry G., Keith R., Salter Venzon D., Samantha I. (2012). Phytochemicals: Health Promotion and Therapeutic Potential. 1st Ed. CRC Press
2. Bagchi D., Francis C. Lau, Ghosh D.K. (2010). Biotechnology in Functional Foods and Nutraceuticals. 4th Ed. CRC press
3. Food Chemistry, O.R.Fennema
4. Food Chemistry, Belitz, Grosch
5. Food Facts & Principles by Shakuntala Manay N & Shadoksharaswamy N, 1996, New Age, World Publishers.

Note: Latest edition of text books may be used.

1	**Course Title**	**Food science**
2	**Course Code**	**NAD 303**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Scientist
8	**Course Objectives**	To provide knowledge to students for preparation of food and the scientific principle behind it**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to CO1**: Summarize on food, food groups and various cooking methods **CO2**: understand the nutritive value and properties of cereals and pulses **CO3**: Have knowledge on composition and nutritive value of milk and various milk products **CO4**: Describe composition, nutritive value, changes during storage and role in cookery of egg, fish and flesh foods **CO5:** Distinguish between fruits and vegetables, its composition, nutritive value and effect of cooking **CO6**: Know the types, properties, nutritive value, properties and changes in cooking of sugars, fats, oils and beverages including processing of coffee, cocoa
10	**Pedagogy of the Course Work**	ClassRoom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

UNIT I

- Cereal and cereal products- cereals and millets. Nutritive value of cereals. Processing of cereals, milling, parboiling, effect of processing on the nutritive value of cereals, dextrinization, gelatinization, gluten formation.
- Pulses and legumes: Production (in brief), processing, use in a variety of preparations, meat substitutes- soya products, textured vegetable protein (TVP) and cost.

UNIT II

- Milk and milk products- composition, classification, quality, processing, nutritive value of milk, butter, curd, butter milk, khoa, cheese and processed milk.
- Vegetables and Fruits – classification, composition and nutritive value. Methods to minimize the loss of nutrients, selection, purchase, storage and availability, cost and use, nutritional aspects of raw and processed vegetables and fruits. Pectins, factors affecting pectin.

UNIT III

- Fats and oils-types, saturated, MUFA, PUFA, sources of fats and oil (animal and vegetables), hydrogenation, invisible fats, use of fat in cookery, factors affecting absorption of fats, smoking point, rancidity, uses, storage cost and nutritional aspects.
- Raising agents- types, constituents, use in cookery and bakery.

UNIT IV

- Convenience foods- role, types, advantages, uses, cost and contribution to diet.
- Tea, coffee, chocolate and cocoa powder, processing, use and nutritional aspects.
- Effect of pH on foods.

13 Suggested Resources for Reading

1. Charley, H.,Food Science, John Wiley& Sons
2. Griswold C., R.H., The Experimental Study of Foods, Houghton Miffin Co.
3. Hughes, O., Introductory Foods, Macmillan & Co
4. Pan and Palamer, Food Theory and Application, John Wiley and Sons
5. Swaminathan, M., Food Science and Experimental Foods, Ganesh &Co.
6. Brain A. Fox and Allan G. Cameron, Food Science, Nutrition and Health, Edward Arnold
7. Mudambi S.R. and S.M. Rao, Food Science, Wiley Eastern LTD, New Delhi, Bangalore, Bombay, Calcutta, Madras, Hyderabad. 8. Devis, B, Food Commodities, Heinman Professional Publishing, Great Britain.

Note: Latest edition of text books may be used.

SEMESTER – III

PW II
Project Work

- "Conduct a cost-benefit analysis of menu offerings in an institutional catering setting (e.g., a hospital or corporate canteen), optimizing meal choices for both nutritional value and budget efficiency, while maintaining food quality and patient satisfaction.

- Submission of project report.

- Viva

Semester IV

1	**Course Title**	**Human Nutrition**
2	**Course Code**	**NAD 401**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Nutritionist, Dietician
8	**Course Objectives**	To provide knowledge regarding digestion and utilization of various nutrients. Apply knowledge of mechanisms of nutrients and their regulation to explain the common deficiency disease related to nutrients
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1** Practically gain knowledge to plan diet for each stage of life according to the guidelines for dietary needs **CO2** Gain knowledge on changes during various stages of growth and development throughout lifecycle **CO3** Thorough understanding of the basis of human nutritional requirements
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Unit I

- Introduction to nutrition, Planning a healthy diet, Food Components, Carbohydrates: Structure, Digestion and Absorption. Disorders related to carbohydrate. Regulation of body temperature and energy metabolism. Glycemic Index and glycemic load. Protein: Structure, Digestion and Absorption. Disorders related to protein. Regulation in food intake, nitrogen balance, protein quality amino acid requirement and imbalances, relation between energy and protein requirement.

Unit II

- Lipids: Structure, Digestion and Absorption. Disorders related to lipids- hyperlipidemia, arthrosclerosis, CHD. Lipids in health management

Unit III

- Vitamins: Classification, functions, RDA, Absorption, deficiency and toxicity of vitamins (Water and fat soluble vitamins)
- Minerals: Classification, functions, RDA, Absorption, deficiency and toxicity of minerals. (Iron, Iodine, Calcium, Phosphorus, Magnesium)

Unit IV

- Lifespan Nutrition- Nutrition assessment and intervention, pregnancy and lactation, lifespan changes in nutritional requirements. Nutrition related diseases- Nutrition and GIT disorders, Nutrition and cardio vascular diseases, Nutrition and diabetes, Nutrition and Evolution

.

13 Suggested Resources for Reading

1. Ross, A. C., Caballero, B., Cousins, R. J., Tucker, K. L., & Ziegler, T. R. (Eds.). (2014). Modern
2. Nutrition in Health and Disease, Wolters Kluwer.
3. Muller, H.G., Fundamentals of Foods and Nutrition. New Delhi, Wiley Eastern Pvt Ltd.
4. Barasi Mary, E., Human Nutrition-A health perspective. Arnold International Student's edition
5. Mehtab S Bamji, Text Book of Human Nutrition, Oxford and IBH Publishing Co. Pvt. Ltd.
6. Joshi S.A., Nutrition and Dietetics with Indian case studies, Mc Graw Hill Education (India) Pvt.L

Note: Latest edition of text books may be used.

1	**Course Title**	**Institutional Food Service and Personnel Management**
2	**Course Code**	**NAD 402**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Service Manager
8	**Course Objectives**	To understand the different kinds of food service units and systems**.** To understand the principles of organization and management. To gain knowledge in various areas of food production**.** To learn the principles of personnel management
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** Understanding food microbiology, sanitation, and hygiene, and how to apply them in food production and service **CO2**. Learning how to design menus and food service techniques, and how to manage food service operations **CO3**. Learning about personnel management, employee benefits, and training **CO4**. Learning how to apply food services technology and operate industry equipment **CO5**. Learning how to communicate appropriately with clients, staff, and management .
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practicals Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

UNIT I

- Types of food service institutions. Styles of food services.
- Planning, writing, types and preparation of menu and menu cards. Menu display.
- Floor planning and layout.

UNIT II

- Principles, functions and tools of management (tangible and intangible).
- Food material management: Meaning definition and importance. Selection, purchase and storage of food. Control in relation to these operations (material planning, budgeting, material standardization, inventory control).
- Store keeping: Definition, objectives, functions, factors underlying successful storekeeping, duties and responsibilities of a storekeeper, purchasing, organization, principles, procedure, systems and quality control

UNIT III

- Personnel management: Recruitment, selection and training of personnel, desirable qualities and responsibilities, work standards, productivity, supervision, performance appraisal, motivation, incentives for effective performance.
- Labour policies and legislation (personnel policies related to salaries, other emoluments, allowances, leave, uniform and other prize benefits.
- Laws affecting food service operations, union and contract negotiations.

UNIT IV

- Financial management: Cost concept, cost control, pricing, budget and accounting, book keeping.
- Types, Selection, purchase, operation, care and maintenance of equipment.

Practical Work

- Standardization of recipes of snacks and a meal for portion and cost.
- Preparation of various types of menu cards.
- Table setting, tray service of tea and snacks of tea and snacks for 20 persons.
- Serving high tea to 20 persons and a meal to 10 persons.
- Preparation and service of packed lunch.
- Planning and preparation of diets for special occasions-birthdays and festivals.
- Maintenance of accounts and record keeping.
- Visit to any two food service institutions (Hospitals, flight kitchen, hotel, restaurant, canteen, industry).
- Prepare an inventory for evaluating staff's personal hygiene.

13 Suggested Resources for Reading

1. Bora, P.M., Food Administration in India. A Study of an Indian State, Ajanta Publishers
2. Boella, M.J., Personnel management in the Hotel and Catering Industry, Bassie and Jenhis, London.
3. Kiesan, J., Controlling, Analyzing Costs in Food Service Operations, MacMillan Publishing Company.
4. Kumar, H.L., Personnel Management in the Hotel and Catering Industry. Metropolitan Book Co., Delhi.
5. Minor, L.J. and Cichy, R.F., Food Service Systems Management. Avi Publishing Co., Connectient.
6. Sethi, M. and Malhan, S., Catering Management-An Integrated Approach. Wiley Eastern Ltd.

Note: Latest edition of text books may be used

1	**Course Title**	**Meal Management**
2	**Course Code**	**NAD 403**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30 hours of lectures for Theory + 60 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (40 THEORY + 10 IA) + (40 PRACTICAL + 10 IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Meal Manager
8	**Course Objectives**	To provide knowledge about basic Principle of meal planning. Exhibit a solid foundation of techniques for food preparation, presentation and service, including competence in diet planning.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1**. Develop critical thinking skill. **CO2**.Learn skills for preparing food **CO3**. Learn to produce nutritious meals **CO4**.Know about sanitation awareness **CO5**.Learn how to plan menus **CO6**.Learn how to purchase food
10	**Pedagogy of the Course Work**	ClassRoom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

UNIT I

- Introduction to meal management: Balanced diet, food groups, food guide and food pyramid, exchange list.
- Basic principles of meal planning, objectives, steps in meal planning, food cost calculation.
- Indian meal patterns: Vegetarian & non-vegetarian.
- Food fads & the faulty food habits.
- Recommended dietary allowances (RDA) for Indians and factors affecting these viz., age, sex, physical activity and physiological state.

UNIT II

- Nutritional and food requirements for adult man and adult woman (moderate worker)
- Nutritional and food requirements for pregnant woman

UNIT III

- Nutritional and food requirements for lactating mother.
- Nutritional and food requirements for an infant :

a) Breast feeding-Advantages and dis advantages, feeding patterns and technique.
b) Infant formula
c) Weaning-Definition, time, importance and steps.
d) Introduction of supplementary food.

- Nutritional and food requirements during early childhood (toddlers and preschoolers)
- Nutritional and food requirements for school going and planning diet with special reference to importance of school lunch and snacks.

UNIT IV

- Nutritional and food requirements for adolescent girl and boy
- Nutritional and food requirements for old age and considerations for planning their diet.

Practical Work

- Plan a day's diet for the following age groups and calculate energy, carbohydrates, protein, fat and one important vitamin or mineral as per requirement for the given age group.

a) Adult man-
b) Adult woman
c) Pregnant woman
d) Lactating woman
e) Infant-weaning food
f) Preschool child
g) School going
h) Adolescent girl
i) Old age

- To prepare any three weaning foods.

13 Suggested Resources for Reading

1. Guthrie, H.A. Introductory Nutrition. 6th ed. Times Mirror/Mosby College Pub. St. Louis.
2. Anderson, L., Nutrition in Health and Disease, J.B. Lippincott co. Philadelphia.
3. Recommended Dietary Intakes for Indians, I.C.M.R.
4. Guthrie, H.A., Introductory Nutrition, Times Mirror/Mosby College Publ.-St. Louis.
5. Worthington, R. & Bonnie S. et al., Nutrition in Pregnancy & Lactation, Times Mirror/Mosby College, St. Louis.

Note: Latest edition of text books may be used

1	**Course Title**	**Food Quality Management**
2	**Course Code**	**NAD 404**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 4 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Quality Manager
8	**Course Objectives**	To understand the different principles of food quality control. To assess the food quality assurance of bakery products.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to CO1.**Learn how to apply safe food handling techniques, water and waste management, and the importance of food laws and regulations **CO2.**Understand the role of food packaging and the importance of nutrition labeling **CO3.**Learn the basic principles of food preservation methods **CO4.**Learn about microbial contamination of food, spoilage, and preservation methods against microbes **CO5.**Learn about foodborne infections and intoxication, sanitation, and hygiene in food industries **CO6**.Learn about different techniques and standard procedures that are followed in industry **CO7.**Learning different certifications required for the food industry
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1.**The syllabus of this paper has been divided into FOUR units.** 2.**Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3.**The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**

4.All questions carry equal marks.

12 Course Content

UNIT I

- Objectives, importance and functions of quality control.
- Principles of food quality control and quality assurance, quality control and assessment in food materials used in baking industry and finished bakery products.

UNIT II

- Total quality management (TQM) - good manufacturing practices, good hygienic practices, good lab practices, general awareness and role of management practices in quality control.
- Microbial quality control: determination of microorganisms in foods by cultural, microscopic, physical, chemical, immunological and bioassay methods

UNIT III

- Food regulations, grade and standards, concepts of Codex Almentarious, HACCP, USFDA, ISO9000 series etc.
- Food laws and standards, Food standards and safety Act: salient provisions and prospects, role of various national and international agencies**.**

UNIT IV

- Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of bakery products

13 Suggested Resources for Reading

1. Alli, I., Food Quality Assurance Principles and Practices, CRC Press.
2. Vasconcellos, J.A., Quality Assurance for the Food Industry. A Practical Approach, CRC Press

Note: Latest edition of text books may be used.

Semester V

1	Course Title	Therapeutic Nutrition I
2	**Course Code**	**NAD 501**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30hours of lectures for Theory + 60 hours of Practical** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Therapeutic Nutritionist
8	**Course Objectives**	• To elucidate the applied concepts of different areas of Foods and Nutrition. • To acquire knowledge about the nutritional needs and concerns of specific age groups/physiological conditions. • To develop the adaptations of normal diet into therapeutic diet**.**
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Understand the etiology, physiology and metabolic anomalies of acute and chronic diseases and patient needs. CO2. Know the effect of various diseases on nutritional and dietary requirements. CO3. Provide and recommend appropriate nutritional care for prevention and treatment of various diseases.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.**

3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**
4. **All questions carry equal marks.**

12 Course Content

UNIT I

• Basic concepts of therapeutic diet.
• Therapeutic adaptations of normal diet
• Principles of diet therapy and therapeutic nutrition for changing needs.

UNIT II

• Modification of normal diet: Liquid diet (clear liquid and full fluid), soft diet, bland diet. • Basic concepts of:
a) Oral feeding
b) Tube feeding
c) Parental nutrition
d) Intravenous feeding

UNIT III

• Obesity-causes, complication, health effects, dietary treatment and other recommendations.
• Underweight- causes, dietary treatment and other recommendations

UNIT IV

• Diet in fevers and infections-metabolism in fever, general dietary considerations, diet in
influenza, typhoid fever and tuberculosis.
• Diet in Diabetes mellitus:
a) Incidence and predisposing factors.
b) Symptoms-types and tests for detection.
c) Metabolism in diabetes.
d) Dietary treatment & meal management.
e) Complication of diabetes.

Practical work

1)Planning of any one diet and calculation of two nutrients important for that particular condition.
2) Planning diets in the following conditions:
a) Obesity
b) Underweight
c) Fever and infections
d) Diabetes mellitus.
3) Planning snacks, desserts and beverages for diabetes.
4) Preparation of clear liquid diet, full fluid diet, soft diet, bland diet.

13 Suggested Resources for Reading

1. Anita, F.P., Clinical Dietetics and Nutrition. Bombay, Oxford Press.
2. Williams, S.R., Nutrition and Diet Therapy. New York, Mosby Mirror Publishing Co.
3. Sri Lakshmi, B., Dietetics. New Delhi, New Age International Pub. Ltd.
4. Malhan, K.N. and Arlim, Krauses Food Nutrition and Diet Therapy. W.B. Saunders

Company, Philadelphia.
5. Passmore, P. and Eastwood, M.A., Human Nutrition and Dietetics. London, ELBS and Co.
Note: Latest edition of textbooks may be used.

1	Course Title	Food Hygiene & Sanitation
2	**Course Code**	**NAD 502**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** **(1 credit = 15 hours of theory learning based on 4 hrs./week)** **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Safety and Hygiene Officer
8	**Course Objectives**	To create awareness about food poisoning, Role of personal hygiene, pest control, cleaning and disinfection and risk of food contamination during packaging, transport, labelling and waste disposal.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Learning about the types of food contamination and the factors that contribute to food borne illness. CO2. Recognize the characteristics of foods that could be potentially hazardous CO3. Learning about the dangers of food borne illness and identifying high-risk populations CO4. Handle food properly to prevent food borne illness CO5. Receive, prepare, serve, and store food products CO6. Identify health problems in food handlers that could threaten food safety CO7. Understand the importance of food safety training in the workplace
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT I

• Concept, significance and interrelationship of health, hygiene and sanitation. Characteristics of good health and factors affecting it.
• Kitchen hygiene and sanitation.
• Hygiene of food handling during procurement, storage, preparation, cooking, serving, holding, cleaning and disposal.

UNIT II

• Personal hygiene of food handlers – Dress, grooming, health and habits.
• Sources of water contamination and its purification at domestic level.
• Causes of air pollution and its ill effects on health.
• Causes, symptoms and control of food borne diseases- diarrhea, dysentery, cholera, typhoid, jaundice.

UNIT III

• Soil impurities, their effect on health and its prevention and control.
• Accidental and bacterial food poisoning. Symptoms and prevention of food poisoning.
• Municipal health services and mobile units. Prevention and control of infectious diseases. Notification, quarantine, segregation and dis- infection.

UNIT IV

• Health services at fairs, festivals, chabils and Langar. Removal and disposal of refuse and excreta.
• Waste product handling: Planning for waste disposal, solid wastes and liquid wastes.
• Control of infestation – Rodent, flies and cockroaches' control, use of pesticides.

13 Suggested Resources for Reading

1. Muller, H.G., Fundamentals of Foods and Nutrition, Wiley Eastern Pvt Ltd.
2. Anderson, L., Nutrition in Health and Disease, Philadelphia, J B Lippincott.
3. Barasi Mary E., Human Nutrition-A health perspective., Arnold International Student's edition, U.K.
4. Calloway, D.H, Nutrition and Health, Philadelphia, Saunders College Publication.
5. S. Roday, Food Hygiene and Sanitation in Food Industry, Tata McGraw Hil
6. Shubhangini A. Joshi, Nutrition and Dietetics, McGraw Hill Education, Pvt Ltd.

Note: Latest edition of textbooks may be used.

1	Course Title	Food Standards and Quality Management
2	**Course Code**	**NAD 503**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** **(1 credit = 15 hours of theory learning based on 4 hrs./week)** **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Food Quality Manager
8	**Course Objectives**	1. To make the students conversant with latest rules and regulation in Food safety as per FSSAI guidelines especially dealing with workplace. 2. To develop the knowledge of sampling procedures and sensory evaluation. 3. To develop the knowledge about grading and marketing standards.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Learn food quality standards. CO2. Develop the skills on the standardization of food products with respect to quality maintain according to universal food standards worldwide. CO3. Understand the principles of sensory evaluation CO4. Develop skills to carry out sensory evaluation of a newly developed product CO5. Understand the terms food adulteration and adulterant.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	**UNIT I** • Importance of quality control and assurance. • Food adulteration: Common food adulterants, consumer awareness, common methods of detecting food adulteration at home and its prevention. • Food laws and regulations-prevention of Food Adulteration Act, Fruit Product Order, Agmark, Essential Commodity Act and Consumer protection and Bureau of Indian Standard, Codex Standards, Hazard analysis and critical control point. **UNIT II** • Sampling procedure. • Application of food standards, sampling and specifications for raw foods and food products for additives, preservatives, coloring matters, antioxidants, emulsifying and stabilizing agents. **UNIT III** • Various methods for the assessment of quality of different foods. • Selection of sensory panel and sensory evaluation of food products. **UNIT IV** • Grading and marketing standards, specifications of finished products. • Market survey of processed foods for quality assurance. • Visit to quality control laboratory and food processing industries. • Market survey of processed foods for quality assurance.
13	**Suggested Resources for Reading**	1. John M. Ryan, Guide to Food Safety and Quality during Transportation: Controls, Standards Practices, Kindle Edition 2. Rees, N., & Watson, D. International Standards for Food Safety, 3. Standard Procedures for Food Safety Management Systems, Bizmanualz, Inc. 4.Moterjami, Y., Food Safety Management: A Practical Guide for the Food Industry, Kindle Edition. **Note: Latest edition of text books may be used.**

1	Course Title	Community Nutrition I
2	**Course Code**	**NAD 504**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Community Nutritionist
8	**Course Objectives**	• To understand the causes/determinants and distribution of nutrition problems in society. • To understand the consequences of nutritional problems. • To be familiar with various approaches to nutrition and health interventions programs and policies.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Understand the ecology of nutritional problems and prevalence in India. CO2. Gain the skills in planning, executing, and evaluating nutrition projects in the community. CO3. Become familiar with various approaches to nutrition and health interventions, programmes, and policies. CO4. Appreciate the national and international contributor towards national improvement in alleviating nutrition problems.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussion, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.**

3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**
4. **All questions carry equal marks.**

12 Course Content

UNIT I

- Concepts of community nutrition- introduction, understanding the major nutritional problems in India.
- Malnutrition, multidisciplinary approach to solving the nutritional problems. Factors affecting distribution of food products, food consumption, storage of food products, measures to overcome malnutrition. Malnutrition and its economic consequences, intervention in malnutrition and government expenditure on intervention. Economics in nutrition (food security, food production, food pricing), economic evaluation of malnutrition.

UNIT II

- Nutritional adaptation and measures to overcome malnutrition-strategies to combat community nutrition problems, diet or food-based strategies, dietary diversification, horticulture intervention, food fortification, germination, supplementation, enrichment, parboiling, GM foods, unconventional foods, green revolution and white revolution.
- Nutrition and infection. Economics of sanitation of food nutrition – loss of food nutrients on contamination with chemical poisons, parasites and microorganisms.

UNIT III

- National and international organization engaged in foods and nutrition activities- National: Role of voluntary agencies and state programmes, contribution of ministries of agriculture and health, ICDS, Nutrient deficiency Control Programs, Supplementary feeding Programs, Food security Programs, ICMR, NIN, CSIR, CFTRI and ICAR. International: FAO, WHO, UNICEF.

UNIT IV

- Review of National Nutrition Programs- rationale, appraisal, impact and costs of improving nutrition situation in India. Food and nutrition security-concept of food security. Factors underlying food and nutrition security.

Practical work

1. Preparation and calculation of following dishes:
• Low-cost Protein rich
• Low-cost iron rich
• Low-cost Calcium rich
• Low-cost vitamin A rich
• Low-cost foods using the principles of mutual supplementation, fortification and enrichment
• Weaning foods
2. Development of nutritious food supplements/ dishes for various vulnerable segments of population.
3. Field visits to:

• Observe the working of nutritional and health-oriented programmes

13 Suggested Resources for Reading

1. McLaren, D.S., Nutrition in the Community, John Wiley and Sons.
2. Jelliffe, D. B., The Assessment of Nutritional Status on the Community-WHO Monograph series No. 53
3. Shukla, P.K., Nutritional Problem of India. Prentice Hall of India Pvt Ltd., New Delhi
4. Ibrahim, G.J., Nutrition in Mother and Children Health, Macmillan.
5. Ritchey, S.J. and Thapar, J., Maternal and Child Nutrition. Harper and Row Publishers.

Note: Latest edition of text books may be used.

1	Course Title	Health, Fitness and Sports Nutrition
2	**Course Code**	**NAD 505**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30hours of lectures for Theory + 60 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Sports Nutritionist
8	**Course Objectives**	1. To learn about the importance of nutrition in sports personnel. 2. To find out the sources of generation of energy for muscle and force generation 3. To know about the ergogenic aids and supplements available in the market.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Explain the principles of physical fitness and nutrition (such as body composition, energy intake, energy expenditure, and the acute and chronic physical changes related to exercise and nutrition) complement each other in helping to develop physiological well-being and overall health. CO2. Explain the principles of fitness and nutrition (such as setting realistic short-term behaviour change goals and the relationship of exercise and diet to stress reduction) complement each other in helping to develop psychological well-being and overall health. CO3. Identify some of the social and cultural influences on food habits and exercise/activity patterns. CO4. Evaluate current nutritional information with regard to its contribution to Health and physical fitness. CO5. Apply the knowledge acquired for planning diet for athletes.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT I

- Sports Nutrition-introduction, evolution and growth of sports nutrition. Fitness and nutritional assessment- - nutrition, exercise, physical fitness and health, concept of nutritional status and body composition. Fitness with reference to sports-flexibility, coordination, equilibrium, speed, strength, endurance. Fuel sources for muscle and exercise metabolism, energy pathways, metabolic response to exercise, metabolic adaptation to exercise training, factors influencing choice of fuels, components of energy expenditure, energy balance**.**

UNIT II

- Macro nutrients in Sports Nutrition-role of carbohydrates before, during and after exercise, carbohydrates loading, protein requirements for exercise, techniques to study protein and amino acid metabolism, effect of protein intake on protein synthesis, amino acids as ergogenic aids, health risks with excessive protein intake, fat as a fuel during exercise, fat supplementation and exercise supplements.

UNIT III

- Micronutrients in Sports Nutrition- role of antioxidants, essential function of vitamins and minerals for athletes, ergogenic effect.
- Water – thermoregulation and exercise in the heat, effect of dehydration in exercise performance, heat illness, fluid guidelines before, during and after exercise.

UNIT IV

- Weight Management and Body Composition- weight management, ideal body weight and composition, weight loss, making weight and rapid weight loss strategies.
- Eating disorders –types, prevalence, risk factors, effect on sports performance, treatment and prevention. Body composition analysis-importance of body composition

Practical Work

1. Development and standardization of sports bars or meal replacement bars.
2. Body fat analysis-learn to use skin fold callipers, bio electrical impedance

analysis technique. Observe DEX analysis.
3. Measurement of blood pressure, heart rate, calculate METs, VO2max
4. Learn about the different types of supplements used by sports persons and role of nutrients listed in athletic performance.
5. Learn to take whole body measurements from a certified fitness trainer using a measuring tape.
6. Guest lecture by a sports nutritionist, fitness trainer, sports physician or physiotherapist on career opportunities.

13 Suggested Resources for Reading

1. Deakin, Burke (2006), 3rd Ed, Clinical Sports Nutrition, McGraw-Hill Australia.
2. Bean, Anita (2006), 5th Ed, Sports Nutrition
3. Bourns, Fred (2002), Essentials of Sports Nutrition, 2nd Ed. John and Wiley.
4. Suzanne Girard Eberle (2000), Endurance Sports Nutrition, Human Kinetics.
5. Benardot, Dan (2000), Advanced Sports Nutrition, Human Kinetics
6. Burke, Louise (2007), Practical Sports Nutrition, Human Kinetics
7. Gleeson, Jeukendrup (2004), Sports Nutrition: An Introduction to Energy Production and Performance, Human Kinetics.

Note: Latest edition of text books may be used.

1	Course Title	Diet Counselling
2	**Course Code**	**NAD 506**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** **(1 credit = 15 hours of theory learning based on 4 hrs./week)** **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Diet Counsellor
8	**Course Objectives**	• To impart skills and confidence in counselling the patients according to their age, sex and physiological condition.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Understanding the diet counselling skills and acquaint them with basic principle. CO2. Determine and translate nutrient needs into menus for individuals and groups across the lifespan, in diverse cultures and religions. CO3. Students will be able to interpret and apply nutrition concepts to evaluate and improve the nutritional health of individuals with medical conditions CO4. Produce oral and written communications for a group education session. CO5. Interview individuals for diet histories and Counsel individuals.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.**

3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**
4. **All questions carry equal marks.**

12 Course Content

UNIT I

- Introduction to term Dietician -definition of dietician, Qualification, classification difference between registered dietician & nutritionist. Role of dietician in hospital. Role of dietician in community, work area of community dietician, role of community dietician.

UNIT II

- Introduction to Nutrition Care Process-definition of nutrition care process, steps of nutrition care process, nutrition assessment-definition, nutrition assessment component, critical thinking.

UNIT III

- Nutrition Diagnosis- nutrition diagnosis domain, intake, clinical, behavioural, environmental. Nutrition diagnosis component, nutrition versus medical diagnosis, medical nutrition therapy.

UNIT IV

- Nutrition Interventions-definition and objectives
  Nutrition monitoring & evaluation.

13 Suggested Resources for Reading

1. Mahan LK and Escott-Stump S (2007): Krause's Food and Nutrition Therapy. 12th Ed. W B Saunders Company, London.
2. Robinson. CH, Lawler MR, Chenoweth WL and Garwick, AE (1986): Normal and Therapeutic Nutrition. 17th Ed., Macmilian Publishing Co.
3. Williams SR (1989): Nutrition & Diet Therapy, 6th Ed. Times Mirror/Mosby College Publishing, St. Louis

Note: Latest edition of textbooks may be used.

M. Marks 50

PWIII

Project Work

- Prepare a diet plan for a patient suffering from multiple health problems (Like blood pressure, diabetes and obesity) and prepare a case study.
- Submitting Project Report.
- Viva

Semester VI

1	Course Title	Therapeutic Nutrition II
2	**Course Code**	**NAD 601**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Therapeutic Nutritionist
8	**Course Objectives**	• To elucidate the applied concepts of different areas of foods and nutrition. • To acquire knowledge about eh nutritional needs and concerns of specific age groups/physiological conditions.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Learn about the nutritional value of different foods and how to customize diet plans for individuals. CO.2 Learn how to prescribe diets for a variety of medical and surgical conditions. CO.3 Learn how to prepare, process, and store food, and how to package it. CO.4 Learn how to participate in community nutritional programs.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1**

(compulsory) of short answer type covering the whole syllabus.

3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.**
4. **All questions carry equal marks.**

12 Course Content

UNIT I

• Diet in gastritis and peptic ulcer-symptoms, clinical findings and dietary modifications. • Diet in disturbances of small intestine and colon.
a) Diarrhoea (child and adult)-classification, modification of diet, fibre, residue, fluid and nutritional adequacy.
b) Constipation-flatulence-dietary considerations.
c) Ulcerative colitis (adults)-symptoms, dietary treatment.
d) Celiac disease, lactose-intolerance, dietary treatment.

UNIT II

• Diet in diseases of the liver, gall bladder and pancreas.
a) Etiology, symptoms and dietary treatment in-jaundice, hepatitis and cirrhosis.
b) Role of alcohol in liver diseases.
c) Dietary treatment in cholecystitis, cholelithiasis and pancreatitis.

- Diet in allergy and skin disturbances: definition, classification, manifestations, common food allergies and test and dietary treatment**.**

UNIT III

• Diet in renal diseases: Basic renal function, symptoms and dietary treatment in acute and chronic glomerulonephritis, nephrosis, renal failure, dialysis. Urinary calculi-causes and treatment, acid and alkali producing and neutral foods.
• Diet in surgical conditions, burns and cancer.

UNIT IV

• Diet in cardiovascular diseases: Role of nutrition in cardiac efficiency, incidence of atherosclerosis, dietary principles, hyperlipidaemia.
• Hypertension-causes and dietary treatment. Sodium restricted diet, level of sodium restriction, sources of sodium.

- Brief knowledge of Dash Diet.
- Food Allergy - Causes, symptoms & dietary management.

Practical Work

1. Planning of any one diet and calculation of two nutrients important for that condition.
2. File

• Planning diets in the following conditions:
a) Diarrhoea
b) Constipation
c) For peptic ulcer
d) For liver disease
e) For cardiovascular diseases.
f) For renal disease.
g) Surgery

13 Suggested Resources for Reading

1. Anita, F.P., Clinical Dietetics and Nutrition. Bombay, Oxford Press.
2. Williams, S.R., Nutrition and Diet Therapy. New York, Mosby Mirror Publishing Co.
3. Sri Lakshmi, B., Dietetics. New Delhi, New Age International Pub. Ltd.
4. Malhan, K.N. and Arlim.Krauses Food Nutrition and Diet Therapy. W.B. Saunders Company, Philadelphia.
5. Passmore, P. and Eastwood, M.A., Human Nutrition and Dietetics. London, ELBS and Co.

Note: Latest edition of textbooks may be used.

1	Course Title	Introduction to Research Methodology and Report Writing
2	**Course Code**	**NAD 602**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** **(1 credit = 15 hours of theory learning based on 4 hrs./week)** **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Research Assistant
8	**Course Objectives**	• The objective of the module is to familiarize the students with basics of research methodology like types of research, data collection, sampling techniques and report writing.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Understand the importance, basic concepts, and ethical issues in research. CO2. Select and define the appropriate research problem. CO3. Explain key components of scientific writing and present the research as well as teaching methodologies
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT I

- Research Methodology: Nature and Scope of Research Methodology. Importance of Research in Important Areas.

UNIT II

- Types of Research: Exploratory Research, Prescriptive Research, Experimental Research.

UNIT III

- Methods of Data Collection: Primary Data-Secondary data. Designing of a Questionnaire, Sampling - Probability sampling and non-probability sampling methods.

UNIT IV

- Business Reports and Proposals: Introduction, Meaning a Report? Steps in Writing a Routine Business Report, Format of Proposals, Four Key Elements in Winning Business Proposals, Case Studies.

13 Suggested Resources for Reading

1. Kothari, C. R., Research Methodology Methods and Techniques, Wiley Eastern Ltd.
2. Rao, S. P. S. S. and Richard, J., An Introduction to Biostatistics – A Manual for Students in Health Sciences, Prentice – Hall of India Pvt. Ltd.
3. Polgar, S. and Thomas, S. S., Introduction to Research in the Health Sciences, Churchill Livingstone Publications
4. Mahajan, B. K., Methods in Biostatistics, Jaypee Brothers Medical Publishers P. Ltd.
5. Gupta, S. P. and Gupta, M. P., Business Statistics, Sultan Chand and Sons.

Note: Latest edition of text books may be used.

1	Course Title	Total Quality Management
2	**Course Code**	**NAD 603**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** **(1 credit = 15 hours of theory learning based on 4 hrs./week)** **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Quality Assurance Manager
8	**Course Objectives**	• The objective of the paper is to provide knowledge to students of the concepts of total quality management and to inculcate among them a concern for quality and customer satisfaction.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Learn the philosophy of TQM and the importance of customer-focused TQM. CO2. Learn how to apply TQM improvement tools to enhance customer satisfaction and improve processes. CO3. Learn how to implement the TQM approach in an organization for continuous quality improvement.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT I

- Introduction: Definition of Quality, Dimensions of Quality, Quality Planning, Quality costs and its Analysis, Basic concepts of Total Quality Management, Principles of TQM.

UNIT II

- Quality Council and Statements, Barriers to TQM Implementation. Quality: Customer Perception and Satisfaction, Customer Complaints, Service Quality, Customer Retention, Continuous Process Improvement – Juran Trilogy, PDSA Cycle, 5S, Kaizen, Supplier Partnership –Selection and Rating

UNIT III

- Business Process Reengineering (BPR). Benchmarking –Introduction, Reasons, Process, Quality Function Deployment (QFD) – House of Quality, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM)

UNIT IV

- Latest Challenges of Quality, Six Sigma Concepts: application. Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System– Elements, Implementation of Quality System, Documentation, Quality Auditing; ISO 14000, Requirements and Benefits.

13 Suggested Resources for Reading

1. Feigenbaum, A.V, Total Quality Management, McGraw-Hill Publication.
2. Oakland, J.S., & Butterworth, Total Quality Management, Heinemann Ltd., Oxford Press.
3. Narayana, V. & Sreenivasan, N. S., Quality Management, Concepts and Tasks, New Age International.

Note: Latest edition of text books may be used.

Seminar I (NAD-604)

1	Course Title	Seminar I
2	**Course Code**	**NAD 604**
3	**Typology of Course**	Seminar
4	**Course Duration/ LP** (Lectures) (Practical)	**120 hours of practical** **(1 credit = 30 hours of practical based on 4 hrs./week)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80PR+20IA)
7	**Job Role**	…..
8	**Course Objectives**	The students will be able to analyse the application of nutrition science to real-world problems and scenarios
9	**Course Outcomes**	The students will be able to • Critically evaluate research studies and articles related to nutrition and dietetics. • Design and deliver effective presentations on nutrition-related topics
10	**Pedagogy of the Course Work**	Presentations, written assignments, case studies, project reports and seminars
11	**Instructions for paper setters and candidates**	**….**
12	**Course Content**	The students will present a seminar on applied topics of Dietetics and Nutrition

1	Course Title	Community Nutrition II
2	**Course Code**	**NAD 605**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Community Nutritionist, Nutrition Counsellor
8	**Course Objectives**	• To impart nutrition education to community.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1.Apply community nutrition services to program planning, intervention and evaluation. CO2. Know about nutritional programs and policies, and how to overcome malnutrition. CO3. Students learn about national, international, and voluntary nutritional organizations that work to combat malnutrition. CO4. Learn how to organize community nutrition education programs using computers. CO5. Learn about the concept of health from both an individual and community perspective. CO6. Learn about common nutritional problems in the community, their causes, symptoms, treatment, and prevention. CO7. Learn about the concept of nutrition security and strategies for combating malnutrition.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	

UNIT I

- Methods of assessing nutritional status-sampling techniques, identification of risk groups.
- Assessing the nutritional problems in the community. Methods for direct assessment- clinical signs, nutritional anthropometry, biochemical estimation and bio-physical methods, diet survey.
- Indirect assessment- nutritionally relevant diseases, ecological parameters, assessment of ecological survey, technique of diet and nutrition survey, food balance sheet and vital statistics**,** Use of growth charts.

UNIT II

- Objectives and operations of feeding programs in the country-pre-school feeding programs, school lunch programs, breast-feeding and its implications, hazards of bottle feeding.
- Weaning foods–Planning, formulating and preparing importance of correct and timely weaning.

UNIT III

- Improvement of nutrition of a community - modern methods to improve nutritional quality of food, food fortification, enrichment and nutrient supplementations. Strategies to improve the quality of food by genetic approaches, clean water, sanitation and street food strategies to improve street food.
- National nutrition policy, Immunization and its importance.

UNIT IV

- Nutrition Education-study of existing daily dietary pattern in relation to socio-economic and psychological aspects, importance of nutrition education for the community, techniques and process of nutrition education. Training workers in nutrition education. Principles of planning, executing monitoring and evaluating the nutrition education programme. Nutrition education themes and messages innutrition and health, antenatal and postnatal care**.**

Practical Work

1. Diet and nutrition surveys

a. To study various anthropometric methods of nutritional status assessment.

b. To study various Biochemical methods of nutritional status assessment.

c. To study about various clinical signs and symptoms used in nutritional assessment. d. To study of various dietary approaches used in nutritional assessment.

e. To study the existing national food security system and report writing

2. Preparation of visual aids for community awareness.

3. Study about various software and applications used in nutritional assessment.

4. Field visit to observe the working of nutrition and health-oriented programs (survey-based result)

13 Suggested Resources for Reading

1.Community Nutrition, Textbook of Public nutrition IGNOU

2. Institutional Food Administration, Mohini Sethi

3. Bloom, W. And Fawcett, D.W.A. Textbook of Histology W.B. Saunders of Company 1968.

4. Guyton, AC, Textbook of Medical Physiology W.B. Saunders & Company.

Note: Latest edition of text books may be used.

1	Course Title	Clinical Assessment and Patient Counselling
2	**Course Code**	**NAD 606**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**30hours of lectures for Theory + 60 hours of practical** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Patient Counsellor
8	**Course Objectives**	To study the details of patient condition, evaluation, counselling to provide appropriate nutritional intervention.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Use clinical skills based on physical examinations, history, and relevant investigations. CO2. Understand the rationale for different therapeutic modalities. CO3. Develop a humane attitude towards patients. CO4. Communicate effectively in a healthcare environment. CO5. Conduct patient assessment and care planning.
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT I

- Health care system-health care service providers (primary, secondary, tertiary), hospital functions, food and dietetics, department and its facilities. Personal hygiene, role in sanitation, safety and first aid, Learning the soft skills and communication.

UNIT II

- Orientation to patient relation, hospital policies and procedures, quality in health care, service and medical quality, TPA and cash management.

UNIT III

- Enteral and parenteral nutrition support, role of immune enhancer, conditionally essential nutrients, immune suppressants and special diets in critical care. Complications of nutritional support -re-feeding syndrome and rehabilitation diets.

UNIT IV

- Critical care nutritional screening and nutritional status assessment of the critically ill nutritional support system and other life saving measures for the critically ill. Enteral nutrition, types of feeds, advantages and disadvantages of easily digestible food. Requirement of various nutrients in relation to disease conditions (hepatic, renal, respiratory). Total Parental Nutrition-importance, long term effect of its use, site of TPN, composition and care

Practical Work

1. Detailed study of a patient case (medical records, prescription, anthropometry, diet, etc.).
2. Evaluation of the patient's condition.
3. Counselling and intervention.
4. Result analysis.
5. File submission and Presentation.

13 Suggested Resources for Reading

1. Mahan LK and Escott -Stump S(2007): Krause's Food and Nutrition Therapy. 12th Ed. WB
Saunders Company, London.
2. Robinson. CH, Lawler MR, Chenoweth WL and Garwick, AE(1986): Normal and Therapeutic Nutrition. 17th Ed., Macmilian Publishing Co.
3. Williams SR (1989): Nutrition & Diet Therapy, 6th Ed. Times Mirror/Mosby College
Publishing, St. Louis.

Note: Latest edition of textbooks may be used.

1	Course Title	Maternal and Child Health and Nutrition
2	**Course Code**	**NAD 607**
3	**Typology of Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** **(1 credit = 15 hours of theory learning based on 4 hrs./week)** **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH+20IA) • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Health Educator
8	**Course Objectives**	• To learn how to evaluate the nutritional needs of mothers and children • To learn how to manage nutritional requirements during pregnancy, lactation, infancy, and childhood • To learn how to provide better guidance and support to patients
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Understand the importance of maternal and child health in the context of nutrition and dietetics CO2. Identify the nutritional needs of pregnant and lactating women, infants, and children CO3. Recognize the role of nutrition in preventing and managing common health problems in mothers and children CO4. Develop skills in assessing and counselling mothers and children on nutrition-related issues CO5. Apply knowledge of maternal and child health to real-world scenarios
10	**Pedagogy of the Course Work**	Classroom Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT-I

- Introduction to Maternal and Child Health
  Overview of maternal and child health, Importance of nutrition in maternal and child health, Global and national perspectives on maternal and child health

UNIT II

- Nutrition in pregnancy – Food and nutrient requirements, physiological changes during pregnancy, developmental stages of the embryo, physiological cost of pregnancy and complications in pregnancy.
- Nutrition in lactation – Food and nutrient requirements, physiology of lactation, composition of breast milk, influence of mother's diet on the quality and quantity of milk production.

UNIT-III

- Nutrition during infancy – Growth and development during infancy, food and nutrient requirements
- Nutrition during preschool age – Food and nutrient requirements, eating habits and behaviour, growth and development and factors inhibiting growth.
- Nutrition for school going children – Food and nutrient requirement, growth pattern, packed lunches, school lunch programmes.

UNIT-IV

- Assessment and Counselling- Assessing nutritional status in mothers and children, Counselling mothers and children on nutrition-related issue, Developing a nutrition care plan for mothers and children, Community Nutrition and Health Promotion
- Community-based approaches to promoting maternal and child health, Health promotion strategies for mothers and children, Role of nutrition in community-based health promotion

13 Suggested Resources for Reading

1. Wardlaw G.M, Hampe J.S, DiSilvestro R.A, Perspectives in Nutrition,6th edition, McGraw Hill, 2004.
2. Chadha R and Mathur P, Nutrition: A Lifecycle Approach. Orient Blackswan New Delhi, 2015.
3. Seth V and Singh K, Diet Planning through the Life Cycle: Part 1Normal Nutrition. A Practical Manual. Elite Publishing House Pvt Ltd. New Delhi,2006.
4. Robinson, Normal and therapeutic nutrition.: Macmillan Pub.Company New York, 2006.
5. Sumati R. Mudambi, M.V. Rajagopal., Fundamental of food, nutrition and diet therapy. New age international publishers, New Delhi, 2015.
6. Srilakshmi B., Dietetics, New age international publishers, New Delhi,2014.

Note: Latest edition of textbooks may be used.